

Roger Richard Demello

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SUMMARY

AI Engineer with experience building LLM applications, RAG systems, and autonomous AI agents. Developed and deployed 3+ production grade AI solutions serving 200+ users using FastAPI, LangChain, vector databases, and cloud technologies. Skilled in backend engineering, machine learning, and AI product development, with a focus on building scalable systems that solve real-world business problems.

WORK EXPERIENCE

AI LifeBOT

Jan 2026 - Present

AI Engineer Intern

- Built and deployed 3+ production-grade LLM applications and autonomous AI agents, serving 200+ users and improving task completion efficiency through tool integration, long-term memory, and goal-driven workflows.
- Architected end-to-end AI pipelines integrating LLMs, RAG, external APIs, and streaming layers, reducing average response latency by 35% and improving system reliability.
- Led rapid prototyping of 5+ AI features, collaborating with cross-functional teams to accelerate product delivery and validation cycles.

CFM, RCOEM

May 2025 - Jul 2025

Machine Learning Intern

- Improved a sleep disorder prediction system using 1,000+ health records and feature engineering techniques, increasing model robustness and generalization.
- Achieved **87% prediction accuracy** through structured feature engineering and statistical analysis of stress levels, sleep duration, and blood pressure.
- Built, trained, and evaluated models in Python with Scikit-learn and Pandas, optimizing end-to-end data workflows to cut development time by 30% and increase model validation accuracy by 25%, leading to faster, more reliable insights.

PROJECTS

Autonomous Executive Email Copilot [GitHub](#)

- Engineered a deterministic RL-style executive inbox simulation environment for benchmarking autonomous email agents across classification, prioritization, and action workflows.
- Developed multi-agent orchestration with FastAPI, vector retrieval, approval gates, and telemetry pipelines for reproducible evaluation and performance monitoring.
- Built policy-driven email management workflows achieving context-aware prioritization through RAG and tool-integrated decision making.

DealSentry - Intelligent Proposal Guard [GitHub](#)

- Built an enterprise AI-powered compliance and risk analysis platform used to automate proposal validation and reduce manual review effort using React, TypeScript, Node.js, PostgreSQL, and Azure OpenAI.
- Designed a rule-based compliance engine with AI-driven risk scoring, document ingestion (DOCX/PDF), approval workflows, audit trails, and contract lifecycle tracking.
- Integrated HubSpot and Gmail APIs, reducing manual compliance review effort and accelerating approval workflows.

SentinelOps - Autonomous Incident Prediction & Prevention Platform [GitHub](#)

- Built an AI-powered platform using FastAPI, React, and anomaly detection models to analyze 10,000+ synthetic telemetry events and automate root-cause analysis.
- Engineered multi-agent workflows for anomaly detection, impact assessment, and remediation recommendations.
- Integrated observability pipelines, audit logging, and Slack/PagerDuty alerts for proactive infrastructure monitoring.

SKILLS

- **Agentic & Generative AI:** RAG Pipelines, Vector Databases, Multi-Agent Systems, Prompt Engineering, Agentic Workflows
- **Machine Learning & AI:** Scikit-learn, NumPy, Pandas, PyTorch, LLMs, Semantic Search
- **Backend & DevOps:** Amazon Web Services, FastAPI, Flask, REST APIs, CI/CD, PostgreSQL, Vercel
- **Programming Languages:** C++, Python, Java, C, SQL
- **Tools & Platforms:** Git, GitHub, Hugging Face, Supabase, LangChain, Docker

EDUCATION

Shri Ramdeobaba College of Engineering and Management

Nov 2022 - May 2026

Bachelor of Technology, Electronics and Communication

- **GPA:** 8.9/10

Shri Ramdeobaba College of Engineering and Management

Jun 2023 - Dec 2025

Minor Degree: Artificial Intelligence and Machine Learning

- **GPA:** 9.6/10

CERTIFICATIONS / AWARDS

- AWS Certified Cloud Practitioner: Amazon Web Services (Issued October 2025)
- Finalist: Paytm x Sarvam x Logitech AI National Hackathon
- 2nd Place: Bytesage AI National Hackathon